

SI 140A.02 Probability & Statistics for EECS, Fall 2026

Homework 1

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Read all the instructions below carefully before you start working on the assignment, and before you make a submission.

- You are required to submit your homework in PDF format generated by **LaTeX**, otherwise you will get no points of this homework.
- You are required to write down all the major steps towards making your conclusions; otherwise you may obtain limited points of the problem.
- Write your homework in English; otherwise you will get no points of this homework.
- Any form of plagiarism will lead to 0 point of this homework.

Problem 1 (20%)

(Story Proof) Let $S(n, k)$ denote the number of ways to partition $\{1, 2, \dots, n\}$ into k nonempty, unlabeled subsets. Give a story proof for each of the following identities.

(a) Prove that

$$S(n+1, k) = S(n, k-1) + kS(n, k).$$

(b) Prove that

$$S(n, k) = \sum_{j=k-1}^{n-1} \binom{n-1}{j} S(j, k-1).$$

Solution

(a) Consider the element $n+1$. There are two possible cases.

First, suppose that $n+1$ is not in a block by itself. We first partition the other n elements into k nonempty blocks, which can be done in $S(n, k)$ ways. We then insert $n+1$ into one of these k blocks. Therefore, this case gives $kS(n, k)$ partitions.

Second, suppose that $n+1$ forms a block by itself. The other n elements must then be partitioned into $k-1$ nonempty blocks, which can be done in $S(n, k-1)$ ways.

Since the two cases are disjoint and include all possibilities,

$$S(n+1, k) = S(n, k-1) + kS(n, k).$$

(b) Fix the element n and consider the elements that are not in the same block as n . There must be at least $k-1$ such elements, since they have to form the other $k-1$ nonempty blocks. There can be at most $n-1$ such elements.

Suppose exactly j elements are not in the same block as n . These j elements can be selected from the other $n-1$ elements in

$$\binom{n-1}{j}$$

ways. They can then be partitioned into $k-1$ nonempty blocks in $S(j, k-1)$ ways. The remaining $n-1-j$ elements must be in the same block as n . Thus, for a fixed j , the number of possible partitions is

$$\binom{n-1}{j} S(j, k-1).$$

Summing over all possible values of j , we obtain

$$S(n, k) = \sum_{j=k-1}^{n-1} \binom{n-1}{j} S(j, k-1).$$

Problem 2 (20%)

A *nonrepeat password* is a nonempty sequence formed from m distinct symbols, where no symbol may appear more than once. The length of the password is ℓ which could be any integer from 1 to m . Suppose that one password is chosen uniformly at random from all nonrepeat passwords.

Let p_m be the probability that the selected password uses all m symbols.

(a) Find an expression for p_m .

(b) Show that

$$p_m = \frac{1}{\sum_{j=0}^{m-1} \frac{1}{j!}} \text{ and } \lim_{m \rightarrow \infty} p_m = \frac{1}{e},$$

where e is the base of the natural logarithm.

Solution

(a) If the password has length ℓ , we first choose ℓ symbols from the m distinct symbols and then arrange them. Thus, there are $\binom{m}{\ell} \ell!$ passwords of length ℓ . Since $1 \leq \ell \leq m$, the total number of nonrepeat passwords is

$$\sum_{\ell=1}^m \binom{m}{\ell} \ell!.$$

There are $m!$ passwords that use all m symbols. Because every password is equally likely,

$$p_m = \frac{m!}{\sum_{\ell=1}^m \binom{m}{\ell} \ell!}.$$

(b) Using $\binom{m}{j} j! = m!/(m-j)!$, we obtain

$$p_m = \frac{m!}{\sum_{j=1}^m \binom{m}{j} j!} = \frac{1}{\sum_{j=1}^m 1/(m-j)!} = \frac{1}{\sum_{j=0}^{m-1} 1/j!}.$$

It remains to show that

$$\lim_{m \rightarrow \infty} \sum_{j=0}^{m-1} \frac{1}{j!} = e.$$

By the binomial theorem,

$$\left(1 + \frac{1}{n}\right)^n = \sum_{k=0}^n \binom{n}{k} \frac{1}{n^k}.$$

For each fixed k ,

$$\binom{n}{k} \frac{1}{n^k} = \frac{1}{k!} \prod_{r=0}^{k-1} \left(1 - \frac{r}{n}\right) \rightarrow \frac{1}{k!} \quad (n \rightarrow \infty).$$

Moreover,

$$0 \leq \binom{n}{k} \frac{1}{n^k} \leq \frac{1}{k!}.$$

The tails of $\sum_{k=0}^{\infty} 1/k!$ tend to zero, so this bound allows us to pass the limit through the sum. Since $\lim_{n \rightarrow \infty} (1 + 1/n)^n = e$, it follows that

$$e = \sum_{k=0}^{\infty} \frac{1}{k!} = \lim_{m \rightarrow \infty} \sum_{j=0}^{m-1} \frac{1}{j!}.$$

Therefore,

$$\lim_{m \rightarrow \infty} p_m = \frac{1}{e}.$$

Problem 3 (20%)

(Geometry Probability) You get a stick and break it randomly into three pieces. Two breaking points are chosen independently and uniformly. What is the probability that you can make a triangle using such three pieces?

Solution

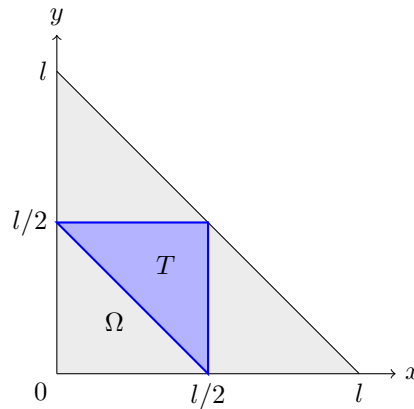
Let the stick have length l , and let two of the three pieces have lengths x and y . The third piece has length $l - x - y$. The sample space is

$$\Omega = \{(x, y) : x > 0, y > 0, x + y < l\}.$$

Because the two breaking points are chosen independently and uniformly, (x, y) is uniformly distributed over this triangular region.

The three pieces form a triangle exactly when each piece is shorter than $l/2$. Thus the valid region is

$$T = \{(x, y) : x < l/2, y < l/2, x + y > l/2\}.$$



Both regions are right triangles. The legs of Ω have length l , while the legs of T have length $l/2$. Therefore,

$$P(\text{the pieces form a triangle}) = \frac{\text{Area}(T)}{\text{Area}(\Omega)} = \frac{\frac{1}{2}(l/2)^2}{\frac{1}{2}l^2} = \boxed{\frac{1}{4}}.$$

Problem 4 (20%)

(Birthday Problem) In the birthday problem, we assumed that all 365 days of the year are equally likely (and excluded February 29). In reality, some days are slightly more likely as birthdays than others. For example, scientists have long struggled to understand why more babies are born 9 months after a holiday. Let $\mathbf{p} = (p_1, p_2, \dots, p_{365})$ be the vector of birthday probabilities, with p_j the probability of being born on the j th day of the year (February 29 is still excluded, with no offense intended to Leap Dayers). The k th elementary symmetric polynomial in the variables $x_1, \dots, x_n$ is defined by

$$e_k(x_1, \dots, x_n) = \sum_{1 \leq j_1 < j_2 < \dots < j_k \leq n} x_{j_1} \dots x_{j_k}, \quad 2 \leq k \leq 365.$$

This just says to add up all of the $\binom{n}{k}$ terms we can get by choosing and multiplying k of the variables.

For example, $e_1(x_1, x_2, x_3) = x_1 + x_2 + x_3$, $e_2(x_1, x_2, x_3) = x_1x_2 + x_1x_3 + x_2x_3$, and $e_3(x_1, x_2, x_3) = x_1x_2x_3$. Now let $k \geq 2$ be the number of people.

- Find a simple expression for the probability that there is at least one birthday match, in terms of $\mathbf{p}$ and an elementary symmetric polynomial.
- Explain intuitively why it makes sense that P (at least one birthday match) is minimized when $p_j = \frac{1}{365}$ for all j , by considering simple and extreme cases.
- The famous arithmetic mean-geometric mean inequality says that for $x, y \geq 0$

$$\frac{x+y}{2} \geq \sqrt{xy}.$$

This inequality follows from adding $4xy$ to both sides of $x^2 - 2xy + y^2 = (x-y)^2 \geq 0$. Define $\mathbf{r} = (r_1, \dots, r_{365})$ by $r_1 = r_2 = (p_1 + p_2)/2$, $r_j = p_j$ for $3 \leq j \leq 365$. Using the arithmetic mean-geometric mean bound and the fact, which you should verify, that

$$e_k(x_1, \dots, x_n) = x_1x_2e_{k-2}(x_3, \dots, x_n) + (x_1 + x_2)e_{k-1}(x_3, \dots, x_n) + e_k(x_3, \dots, x_n)$$

show that

$$P(\text{at least one birthday match} \mid \mathbf{p}) \geq P(\text{at least one birthday match} \mid \mathbf{r})$$

with strict inequality if $\mathbf{p} \neq \mathbf{r}$, where the given $\mathbf{r}$ notation means that the birthday probabilities are given by $\mathbf{r}$. Using this, show that the value of $\mathbf{p}$ that minimizes the probability of at least one birthday match is given by $p_j = \frac{1}{365}$ for all j .

Solution

- (a) To have no birthday match, the k people must have k distinct birthdays. For each choice of k distinct days, there are $k!$ ways to assign them to the k people. Hence

$$P(\text{at least one match} \mid \mathbf{p}) = 1 - k!e_k(p_1, \dots, p_{365}).$$

- (b) Consider two simple cases. When $k = 2$, the probability of a match is $\sum_{j=1}^{365} p_j^2$. Averaging two unequal probabilities decreases the sum of their squares, so the uniform distribution gives the smallest match probability.

When $k = 365$, the probability of no match is $365! \prod_{j=1}^{365} p_j$. By the arithmetic mean-geometric mean inequality, this product is largest when all the p_j 's are equal. At the other extreme, if everyone is born on the same day, a match is certain. These cases suggest that making birthdays more evenly distributed reduces the chance of a match.

- (c) In the definition of e_k , each term contains either both x_1, x_2 , exactly one of them, or neither of them. These three disjoint cases give

$$\begin{aligned} e_k(x_1, \dots, x_n) &= x_1 x_2 e_{k-2}(x_3, \dots, x_n) \\ &\quad + (x_1 + x_2) e_{k-1}(x_3, \dots, x_n) + e_k(x_3, \dots, x_n). \end{aligned}$$

Averaging p_1 and p_2 leaves their sum unchanged, so only the first term changes. In fact,

$$e_k(\mathbf{r}) - e_k(\mathbf{p}) = \left(\frac{(p_1 + p_2)^2}{4} - p_1 p_2 \right) e_{k-2}(p_3, \dots, p_{365}) = \frac{(p_1 - p_2)^2}{4} e_{k-2}(p_3, \dots, p_{365}) \geq 0.$$

Therefore,

$$P(\text{at least one match} \mid \mathbf{p}) \geq P(\text{at least one match} \mid \mathbf{r}).$$

The inequality is strict if $p_1 \neq p_2$ and $e_{k-2}(p_3, \dots, p_{365}) > 0$.

Finally, let $2 \leq k \leq 365$. Since e_k is continuous on the compact probability simplex, it attains a maximum. The uniform vector gives $e_k = \binom{365}{k} (1/365)^k > 0$, whereas any vector with fewer than k positive entries gives $e_k = 0$. Thus a maximizing vector must have at least k positive entries. After removing any two entries, at least $k - 2$ positive entries remain, so e_{k-2} of the remaining entries is positive (with the convention $e_0 = 1$). Assuming that two of its entries are unequal, averaging them would strictly increase e_k by the argument above, which leads to a contradiction. Thus all entries must be equal. Consequently, the uniform vector minimizes the probability of at least one birthday match.

Problem 5 (20%)

A card collection contains five different types of cards. Each independent draw produces card type i with probability p_i , where

$$(p_1, p_2, p_3, p_4, p_5) = (0.05, 0.15, 0.20, 0.25, 0.35).$$

Let Q_n denote the probability of collecting all five card types after n independent draws. $n \geq 1$.

For each $i \in \{1, \dots, 5\}$, define

$$E_i = \{\text{card type } i \text{ does not appear in the first } n \text{ draws}\}.$$

For any index set

$$I \subseteq \{1, \dots, 5\},$$

define

$$E_I = \bigcap_{i \in I} E_i.$$

Thus, E_I is the event that none of the card types whose indices belong to I appears in the first n draws. By convention,

$$E_\emptyset = \Omega,$$

where Ω denotes the entire sample space.

- (a) Use the inclusion-exclusion principle to prove that

$$Q_n = \sum_{I \subseteq \{1, \dots, 5\}} (-1)^{|I|} \left(1 - \sum_{i \in I} p_i\right)^n,$$

where $|I|$ denotes the number of elements in I .

- (b) Find the smallest integer n such that $Q_n \geq 0.90$. You can write some code to calculate it. Include the screenshot of your code and output in the terminal which can support your answer.
- (c) Find the smallest n for which the probability of collecting all five types is at least 0.90 in the uniform case. (No need to include the code) Write code to plot Q_n as a function of n for both equal and unequal p_i , along with their corresponding probabilities. (You should include the graph.) Compare Q_n with the corresponding probability between the two cases.

Solution

- (a) Collecting all five card types is the complement of the event $E_1 \cup \dots \cup E_5$, which means that at least one type is missing. For any $I \subseteq \{1, \dots, 5\}$, each draw avoids every type in I with probability $1 - \sum_{i \in I} p_i$. Since the draws are independent,

$$P(E_I) = \left(1 - \sum_{i \in I} p_i\right)^n.$$

Applying the inclusion-exclusion principle to $E_1 \cup \dots \cup E_5$ and then taking the complement gives

$$Q_n = \sum_{I \subseteq \{1, \dots, 5\}} (-1)^{|I|} \left(1 - \sum_{i \in I} p_i\right)^n.$$

The term for $I = \emptyset$ equals 1, as required.

- (b) Evaluating the inclusion-exclusion formula gives $Q_{45} \approx 0.899893 < 0.90$ and $Q_{46} \approx 0.904965 \geq 0.90$. Since Q_n is nondecreasing in n , the smallest integer satisfying $Q_n \geq 0.90$ is $\boxed{46}$. The code and its terminal output are shown below. Both views are cropped from the same screenshot.

```

from itertools import combinations
import numpy as np

prob = np.array([0.05, 0.15, 0.20, 0.25, 0.35])
n = 1

def solve(x, y):
    result = 0.0
    for size in range(len(y) + 1):
        for subset in combinations(range(len(y)), size):
            result += (-1) ** size * (1 - sum(y[i] for i in subset)) ** x
    return result

while True:
    if solve(n, prob) >= 0.90:
        print(f"The smallest integer n such that Q_n >= 0.90 is {n}, with Q_n = {solve(n, prob):.12f}")
        break

    else:
        print(f"Q_{n} is {solve(n, prob):.12f}")
        n += 1
        continue

```

问题 21 输出 调试控制台 终端 端口

```

(base) PS E:\local-repository> & C:\ProgramData\anaconda3\python.exe e:/local-repository/SI140/hm1.py
Q_28 is 0.751658122510
Q_29 is 0.765119662289
Q_30 is 0.777753236875
Q_31 is 0.789622938781
Q_32 is 0.800786415597
Q_33 is 0.811295628274
Q_34 is 0.821197521422
Q_35 is 0.830534612819
Q_36 is 0.839345509972
Q_37 is 0.847665361659
Q_38 is 0.855526252027
Q_39 is 0.862957544257
Q_40 is 0.869986180202
Q_41 is 0.876636941679
Q_42 is 0.882932678502
Q_43 is 0.888894507680
Q_44 is 0.894541987675
Q_45 is 0.899893271078
The smallest integer n such that Q_n >= 0.90 is 46, with Q_n = 0.904965238641
(base) PS E:\local-repository>

```


(c) In the uniform case, each card type has probability $1/5$, so inclusion-exclusion simplifies to

$$Q_n^{\text{uniform}} = \sum_{j=0}^5 (-1)^j \binom{5}{j} \left(1 - \frac{j}{5}\right)^n.$$

The computed values are $Q_{17}^{\text{uniform}} \approx 0.889101 < 0.90$ and $Q_{18}^{\text{uniform}} \approx 0.910943 \geq 0.90$. Hence the smallest n in the uniform case is 18. At $n = 18$, the unequal case has $Q_{18} \approx 0.550869$, much lower than the uniform case. The graph shows that the uniform probabilities reach the 0.90 target 28 draws earlier.

